15/40

VIRTUAL SCHOOL VICTORIA

2026

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ALGORITHMICS UNIT 3

SAC 1: Abstract Data Types (Weeks 1 to 4)

Outcome 1

Date of Completion: 2-6 March 2026

Reading Time: 5 minutes
Writing time: 55 minutes
TOTAL (60 minutes)
QUESTION AND ANSWER BOOK

Type	*Number of questions*	*Number of questions to be answered*	*Number of marks*
Short/Extended Response	9	9	40
		Total	40

Materials supplied
- Question and answer book of 10 pages

Materials permitted
- Pens/Stationary

No calculator or reference material permitted.

Instructions
- Write your **name** in the space provided above on this page.
- All written responses must be in English, point form is preferred.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices.

1.

a. Define a tree.

A connected graph with no cycles.



(1 mark)

b. What possible numbers of edges could a tree with 10 vertices have?

9 edges

(1 mark)

Q. Give one similarity and one difference between a set and a dictionary ADT.

A dictionary and a set are both unordered.
A dictionary can have duplicate ~~values~~ values, but a set cannot.

(2 marks)

2.

(a) The following items are added to a minimum priority queue. The priority is given in brackets.

Added 1st: Alpha (2)

Added 2nd: Bravo (1)

Added 3rd: Charlie (1)

Added 4th: Delta (2)

D² A² C¹ B¹

Write down the order in which they are dequeued, from first to last.

Bravo, Charlie, Alpha, and lastly Delta.

(2 marks)

(b) Write down the signature for a method to add an element to a priority queue.

addElement: priorityQueue × element × int → priorityQueue

'int' to assign a priority as an integer, which can work in the context of this question too.

(2 marks)

3. Consider the following pseudocode:

```
a = 2
b = 10
if a > b:
  c = 0
  d = b
  while a > d:
    d = d * b
    c = c + 1
  return c
else:
  swap values of a and b
  c = 0
  d = b
  while a > d:
    d = d * b
    c = c + 1
  return c
```

modularised code.

```
a = 2
b = 10
c = 0
define function
    d = b * b
    c = c + 1
    return c
if a > b:
    function()
else:
    swap values of a and b
    function()
```

a = 10, b = 2, c = 0, d = 2
d = 4, c = 1

a. Show what the code returns using the values given.

c = 1

(2 marks)

b. By annotating the code, show how modularisation can be used to improve the code, stating clearly one advantage of your approach.

It is more efficient to store the actions we want done in a function, which we can keep referencing instead of hard coding it each time, which also saves time and energy when writing code, and also helps to keep any mistakes from accidently occurring. We also don't need 'd=b', as shown above.

(2 marks)

4. Find the values of A, B and C which make the following Boolean expression False.

`{not (A and B)} or (A and C)`

~~1 = A, B, C is true~~
~~0 = A, B, C is false~~

(A' and B') or (A and C)

C = 0, False

A = 0 when C = 0 or 1 = false

A = 0, B = 1 → False

A = 1, B = 0 → False

A = 1, B = 1 → False

1 = true

0 = false

ans?

(2 marks)

5. Bruce is constructing a model to organise his music collection. The table below shows the kind of information that needs to be stored:

`Song name, Artist Name, Genre, Play Count, Length`

He wants to be able to quickly look up songs either by artist or by genre. Initially he does this using a list of dictionaries, where each song is represented by a dictionary.

```
song_list = [
...
{name: "Taking The Queen", artist="Bruce Dickinson",
genre="Heavy Metal", play count=15, length="4.48"}
...
]
```

(a) Explain two distinct reasons that this combination of ADTs is suboptimal.

Because it would be faster to use a dictionary of lists, as it's faster to just attribute a song name to a key. It is also inconvenient to have so many dictionaries, as ~~it is easier to model this using a singular dictionary.~~ ~~This combination in the~~ will make it trickier to search by genre or artist, as you have to access the list first. (3 marks)

X

✓

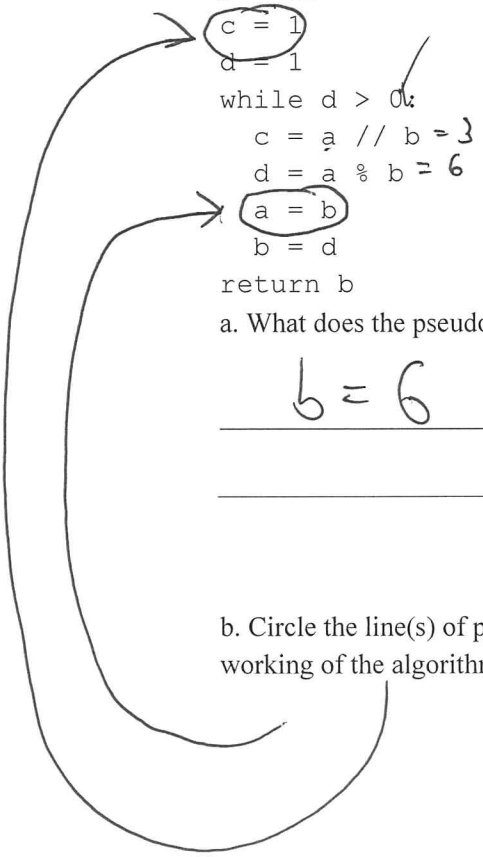
(b) Construct your own combinations of ADTs to solve his problem. Clearly explain why your solution allows efficient lookup of songs by both artist and genre.

I would use a dictionary of lists, with the song name being the key, and value being the list where the song info is. This is efficient because it is easy to just enter in an artist or genre, and our model will show all keys that have values corresponding with the artist/value we entered.

(5 marks)

6. Consider the following pseudocode

```
a = 30
b = 12
c = 1
d = 1
while d > 0:
    c = a // b = 3
    d = a % b = 6
    a = b
    b = d
return b
```



a. What does the pseudocode return when run with the given values?

b = 6

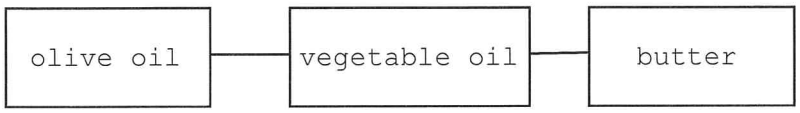
(2 marks)

b. Circle the line(s) of pseudocode which could be removed without changing the overall working of the algorithm.

(2 marks)

7. Jayden wants to create an AI system which can help you to complete recipes with missing ingredients by indicating which ingredients can be substituted for others.

He starts with this graph, indicating that butter or olive oil can both be used instead of vegetable oil.



a. This erroneously leads to somebody trying to pour butter over a salad instead of olive oil. Explain why this happened, and how the graph can be changed to avoid this problem.

Because the graph does not specify ~~which order (AND) direction~~ ~~what~~ in what situation we should swap them in.

(2 marks)

b. Under what situation might you decide to use weights on the edges of the graphs?

To indicate the quantity of olive oil or butter we should add compared to ~~vegta~~ vegetable oil.

(2 marks)

c. Explain how we could further modify the model to indicate that, for example, olive oil can be substituted for butter in savoury dishes but not sweet ones.

We could turn the model into a decision tree so we can ask whether our dish is sweet or savoury. If sweet, don't add olive oil. If savory, you can add it.

(2 marks)



8. You are to use ADTs to construct a model which could be used to find a solution to the 'jealous husbands' problem with four couples: There are four married couples on one side of a river. They have a boat which carries two people. At least one person must be in the boat to cross the river. The husbands are all jealous, such that they will not allow their wives to be in the presence of another man, unless they themselves are present. That means that if a man's wife is on one side of the river, then either he must be on the same side, or there must be no other men on that side.

a. Explain how each state of the problem can be represented using ADTs.

We could use a connected bipartite graph, ~~to represent all the couples on one side of the river~~ with a dictionary to represent each person. Key = name value, list with info (gender, who they are married to).

(2 marks)

b. Show how your combination can be used to represent the initial state.

We can show the nodes on the left to represent all 4 couples on one side of the river at first.

(1 mark)

c. Show two further nodes which can be achieved from the initial state within the constraints of the problem.

(2 marks)

9. Construct a decision tree to distinguish the following animals using only Yes-No questions, using the properties of the animals. Your tree must have a maximum of three nodes, and asks only two different questions. (i.e. two nodes will ask the same question). You must use correct notation.

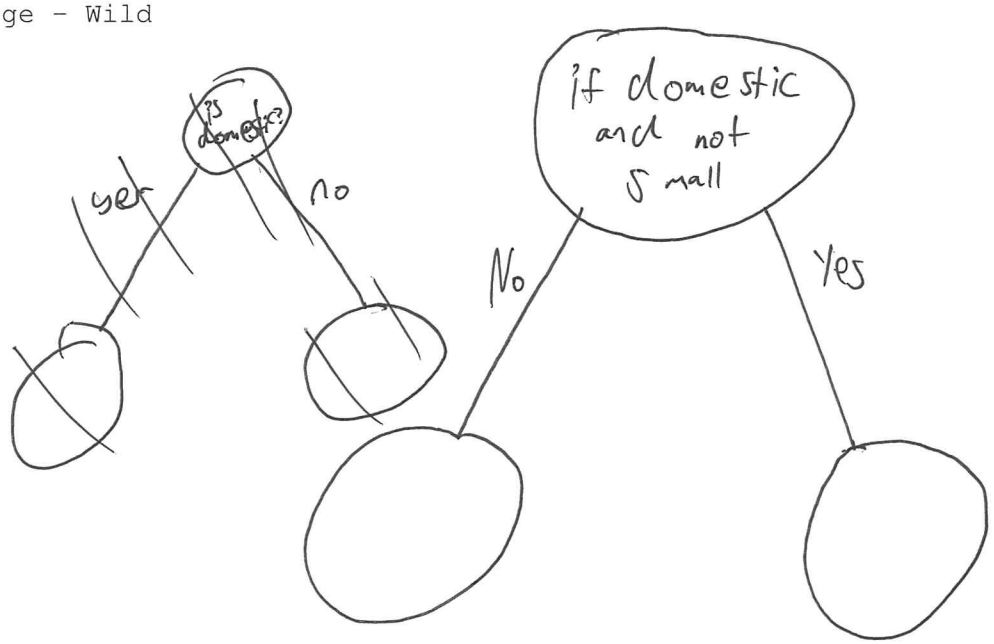
Cat - Small - Domestic

Dog - Medium - Domestic

Echidna - Small - Wild

Elephant - Large - Wild

if domestic and not small = medium = dog
if wild and not small = elephant



9 (3 marks)

STUDENT DECLARATION OF AUTHENTICITY

This SAC is my own work and has been completed under the stated requirements.

Student Signature: [signature] Date: 05/03/26

Supervisor comments: